

InteractionKit

Testing whether corrective interfaces must match how AI communication fails

Does matching an evidence-based corrective interface to an AI answer's communication-failure family improve appropriate reliance relative to an equally formatted, truthful but failure-mismatched correction?

Why this project

AI advice can fail while sounding numerically precise or while omitting the conditions under which a recommendation applies. Existing uncertainty interfaces often ask whether adding confidence, evidence, or explanations helps on average. This project tests a narrower causal principle: corrective information should expose the information missing from the specific communication failure.

Causal structure

Failure family	Unsupported numerical precision	Omitted decision boundary
Corrective card	Numerical warrant	Boundary condition
Key contrast	Failure-matched card	Truthful but failure-mismatched card

Planned design

Within participant: failure family x answer accuracy x displayed confidence x intervention type. Across scenarios: objective evidence support. Each participant completes 16 trials, including eight matched and eight mismatched corrections. The planned pool contains 24 independently grounded exercise-and-health scenarios.

Primary outcome: probability movement toward the correct final decision. Secondary confirmatory outcome: final decision accuracy. Participants and scenarios are treated as crossed sources of variation.

What exists - and what does not

Verified now	Not yet evidence
Public v1.0.0 tag at commit 91bda82; three typed primitives; composition checks; generated schemas; structured JSONL; seven contract tests; TypeScript and production build; clean-install reproduction; frozen Study 2 design and simulation code.	No completed Study 2 scenarios or cards; no independent material ratings; no ethics approval or public registration; no participant data; no causal, construct-validity, cross-domain, or cross-lab result.

Recruitment gates

1. Independently adjudicate 24 scenarios and objective evidence support.
2. Blind-code failure-family purity; remove mixed-failure variants.
3. Pretest card relevance, leakage, length, credibility, and comprehension.
4. Verify the seeded allocation, full-rank matrix, logging, and recovery path.
5. Pass exact-schedule simulation: $\geq 80\%$ power plus Type I error, coverage, and wrong-sign gates.
6. Obtain institutional ethics approval and freeze a public preregistration.

Sample-size boundary

The frozen document uses 240 analyzable participants as a provisional target, not a licensed recruitment claim. It becomes binding only if the exact-schedule post-pilot simulation passes. If scenario heterogeneity is the binding constraint, adding participants alone is rejected; the scenario pool or manipulation must be redesigned.

FALSIFICATION: If failure-matched cards do not improve probability movement toward the correct decision relative to mismatched cards, the proposed failure-contingent interaction principle is not supported.

Review path: maxguo.dev/research/building-trustworthy-ai/interactionkit/research-brief/ | Code: github.com/GBX-Max1220/InteractionKit | Contact: gbx1220max@gmail.com